Supplemental Tables

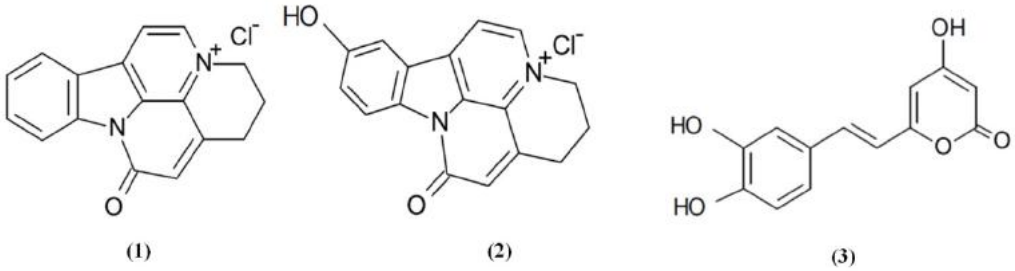
Supplemental Table 1. Variety of mushroom species that were found to have beneficial effects in neuronal diseases

Mushroom	Common name	Taxonomy			Description	Bioactivity	References
		Class	Order	Family			
Hericium erinaceus (Bull.: Fr.) Pers.	Lion's mane, monkey's head, hedgehog mushroom, satyr's beard, pom pom, bearded tooth, and Yamabushitake	Agaricomycetes	Russulales	Hericiaceae	Large, white to creamy, irregular fruiting body with icicle-like projections.	Neurite outgrowth activity; Inhibition of β-amyloid; Neuroprotection	Mori *et al*.(2011) Mori *et al*. (2009) Wong *et al.* (2007) Kawagishi and Zhuang' (2008)
Ganoderma lucidum (Fr) P. Karst	Lingzhi, reishi	Agaricomycetes	Polyporales	Ganodermataceae	Corky, flat, red-varnished, kidney-shaped cap	Neurite outgrowth activity; Inhibition of β-amyloid; Neuroprotection ;	Lai *et al.* (2008) Cheung *et al*. (2000) Chen *et al*. (2007)
Lignosus rhinocerus (Cooke) Ryvarden	Tiger's milk mushroom	Agaricomycetes	Polyporales	Polyporaceae	Posses underground tuber-like sclerotium, solitary fruiting body	Neurite outgrowth activity	Eik *et al.* (2012)
Grifola frondosa (Dicks.: Fr.) S.F. Gray	Hen of the woods, dancing mushroom, Maitake	Agaricomycetes	Polyporales	Fomitopsidaceae	Manifold curled or spoon-shaped, grayish-brown caps, has tuber-like sclerotium	Neurite outgrowth activity	Nishina *et al.* (2006)
Mycoleptodonoides aitchisonii (Berk.) Maas Geest.	Breech oyster mushroom	Agaricomycetes	Polyporales	Mycoleptodonoides	Edible mushroom mainly found in the Kashmir region in India and in Japan. White to yellow cap with a smooth surface, and short stem. Savoury taste.	Neurite outgrowth activity; Neuroprotection	Okuyama *et al*. (2004a, 2004b) Okuyama *et al*. (2012) Kokubo *et al.* (2011) Choi et *al.* (2009)
Antrodia camphorata (M. Zang & C.H. Su) Sheng H. Wu, Ryvarden & T.T. Chang	Niuchangchih	Agaricomycetes	Polyporales	Fomitopsidaceae	A unique Taiwan mushroom, grows only on the inner cavity of the endemic tree species, *Cinnamomum*	Inhibition of β-amyloid; Neuroprotection	Lu *et al*. (2006) Wang *et al*. (2012)

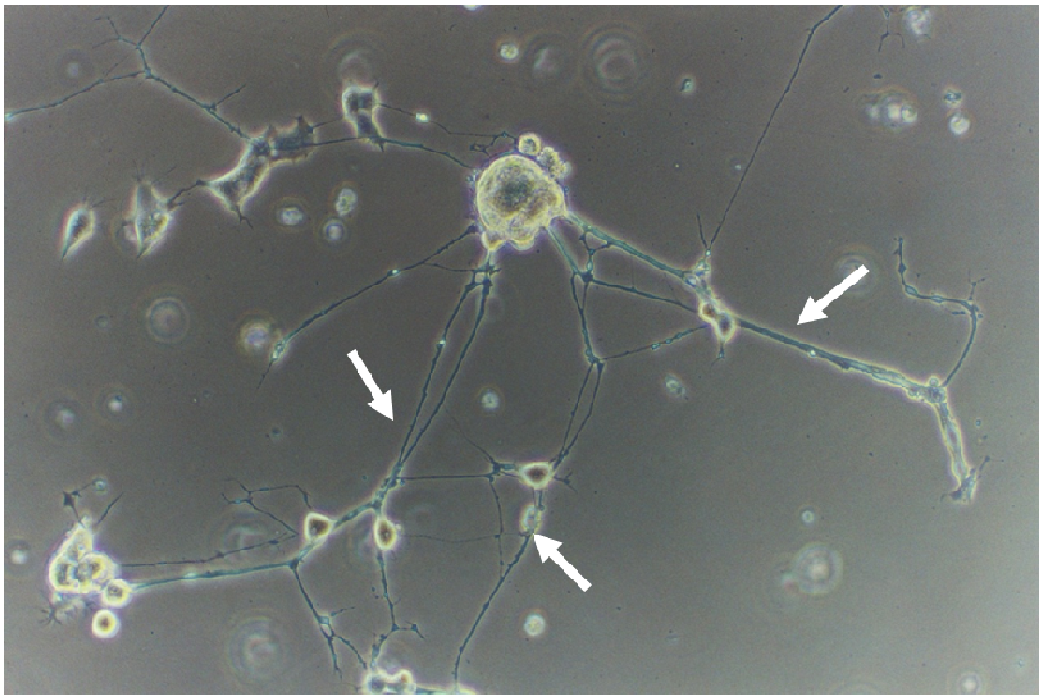
					kanehirai		
Cordyceps militaris (L.:Fr.) Link	Caterpillar fungi, Dongchongxiacao, winter worm summer grass	Sordariomycetes	Hypocreales	Clavicipitaceae	Parasitic to larvae and pupae of moths and butterflies, club-sized fruiting body	Neurite outgrowth activity	Lee *et al.* (2011)
Phellinus linteus (Berk. & M.A. Curtis) Teng	Songgen (Chinese), meshimakobu (Japanese), Sang Hwang (Korean)	Agaricomycetes	Hymenochaetales	Hymenochaetaceae	Grows on mulberry trees, brown to yellow cap	Inhibition of BACE1	Dai *et al*. (2010) Park *et al.* (2004a, 2004b) Sheean *et al.* (2012)
Inonotus obliquus (Ach. ex Pers.) Pilát	Chaga mushroom	Agaricomycetes	Hymenochaetales	Hymenochaetaceae	Parasitic on birch tree, melanin-rich, black	Neuroprotection	Jung *et al.* (2008) Kim (2011) Giridharan *et al.* (2011) Nakajima *et al.* (2009) Chen *et al*. (2010)
Sarcodon cyrneus Maas Geest.	Bitter tooth	Agaricomycetes	Thelephorales	Bankeraceae	Distinctly bitter flavour; *"cyrneus"* for its Mediterranean habitat	Neurite outgrowth activity	Marcotullio *et al.* (2007) Obara *et al.* (2007) Marcotullio *et al.* (2006)
Sarcodon scabrosus(Fr.) P. Karst	Bitter tooth	Agaricomycetes	Thelephorales	Bankeraceae	Brown cap with scales, bitter taste	Neurite outgrowth activity	Ohta *et al.* (1998) Kita *et al*. (1998); Waters *et al.* (2005) Obara *et al.* (2001) Shi *et al*. (2011) Obara *et al.* (1999)
Leccinum extremiorientale (Lar. N. Vassiljeva) Singer	Far-eastern Scaber stalk	Agaricomycetes	Boletales	Boletaceae	Red brown areolate cap	Neuroprotection	Choi *et al.* (2011)
Pleurotus giganteus (Berk.) Karunarathna & K.D. Hyde	Zhudugu, cow's stomach mushroom, morning glory mushroom	Agaricomycetes	Agaricales	Agaricale	Large, thick, and convex cap	Neurite outgrowth activity	Phan *et al.* (2012)
Tremella fuciformis Berk	Yin er, white jelly fungus, silver Ear mushroom, snow mushroom	Tremellomycetes	Tremellales	Tremellaceae	White, frond-like, gelatinous fruiting body	Neurite outgrowth activity	Kim *et al.* (2007) Park *et al*. (2012)

Stropharia rugosoannulata Farlow	Wine cap, burgundy cap, wine cap stropharia	Agaricomycetes	Agaricales	Strophariaceae	Red, brown to tan cap; bell-shaped when young but flat when old	Neuroprotection	Wu *et al*. (2011) Wu *et al*. (2012)
Armillaria mellea (Vahl) P. Kumm.	Honey mushroom	Agaricomycetes	Agaricales	Physalacriaceae	Pathogenic to plant, fruiting bodies found in clumps, sticky, yellow to brown cap	Neuroprotection	Watanabe *et al.* (1990) Gao *et al*. (2009) Muszyńska *et al.* (2011)
Cortinarius infractus (Pers.) Fr.	Bitter webcap	Agaricomycetes	Agaricales	Cortinariaceae	Greyish brown cap, bitter taste	Inhibition of AChe activity	Brondz *et al.* (2007) Geissler *et al.* (2010)
Dictyophora indusiata (Vent.) Desv.	"Queen of the mushrooms'', bamboo mushroom, Kinugasatake (Japanese)	Agaricomycetes	Phallales	Phallaceae	Also called *Phallus indusiata*, conical to bell-shaped cap covered with a greenish-brown spore and slime	Neurite outgrowth activity	Kawagishi *et al.* (1997)
Daldinia concentrica (Bolton) Ces. & De Not.	King Alfred's cake, cramp balls, coal fungus	Sordariomycetes	Xylariales	Xylariaceae	Ball-shaped, coal-black fruiting body	Neuroprotection	Quang *et al.* (2002)
Tricholoma sp. (Fr.) Staude	Matsutake (Japanese)	Agaricomycetes	Agaricales	Tricholomataceae	Mycorrhizal, gilled cap, fleshy stems	Neurite outgrowth activity; Inhibition of AChe activity	Tel *et al*. (2011) Tsukamoto *et al.* (2003)
Termitomyces albuminosus (Berk.) R. Heim	Jizong (Chinese)	Agaricomycetes	Agaricales	Lyophyllaceae	Long stem, edible. Emerge from termite nest.	Neurite outgrowth activity	Shi *et al*. (2012)
Termitomyces titanicus Pegler & Piearce	-	Agaricomycetes	Agaricales	Lyophyllaceae	Cap diameter of up to1m; the largest edible mushroom in the world according to Guinness Book of Records.	Neuroprotection	Choi *et al.* (2010)

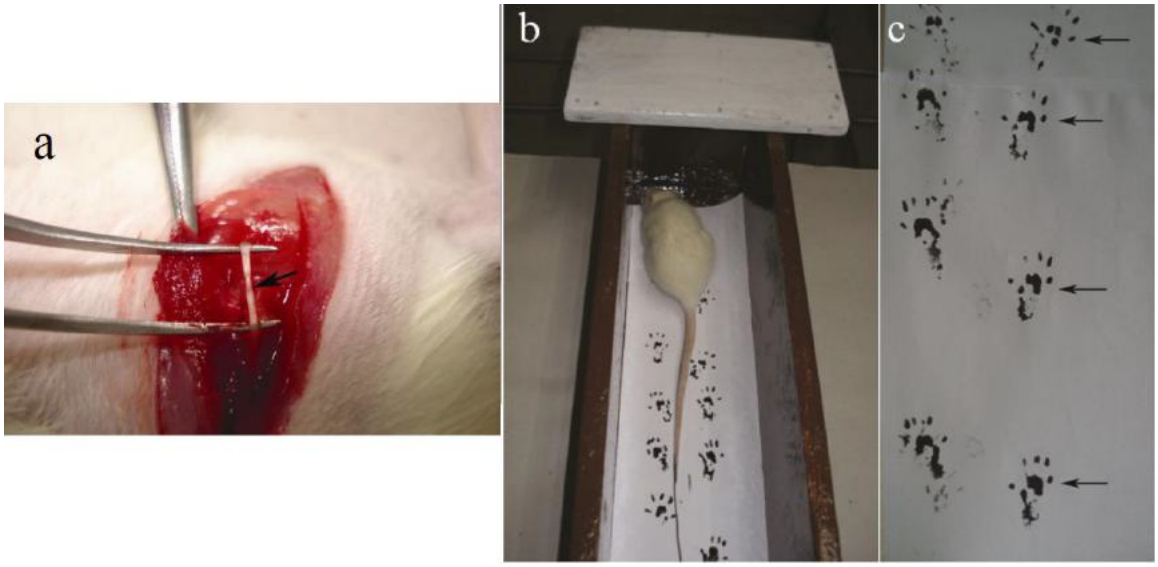
Supplemental Figures



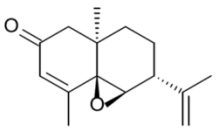
Supplemental Figure 1. Infractopicrin **(1)** and 10-hydroxy-infractopicrin **(2)** isolated from *Cortinarius infractus;* and Hispidin **(3)** from *Phellinus linteus.*



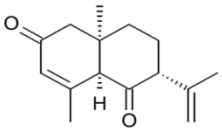
Supplemental Figure 2. Neurite formation of NG108-15 cells induced by aqueous extracts of *H. erinaceus* fruiting body. Arrows indicate neurite extension.



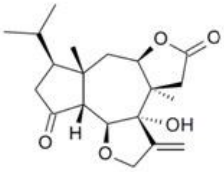
Supplemental Figure 3. (a) A complete crush of peroneal nerve is confirmed by presence of a translucent band (as indicated by an arrow) across the nerve. (b) Walking track apparatus. (c) Walking tracks of footprints after 4 days of peroneal nerve crush injury. Arrows indicate footprints of the operated limb. Foot prints in *H. erinaceus* aqueous extract group demonstrated toe-spreading and clear footprints of the operated limbs on the walking tracks.



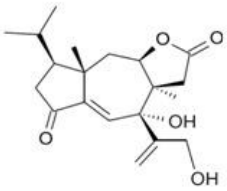
(43)



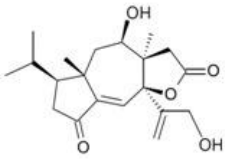
(44)



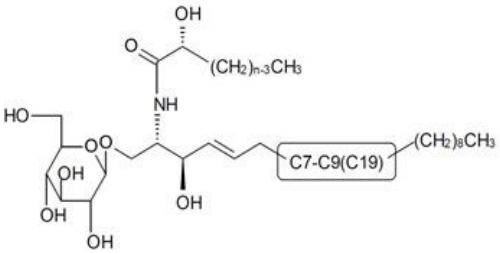
(45)



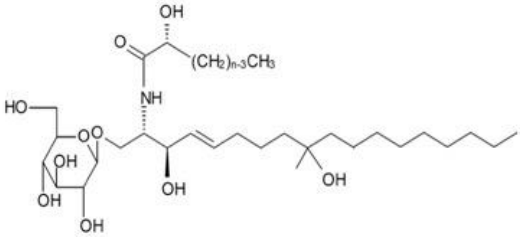
(46)



(47)

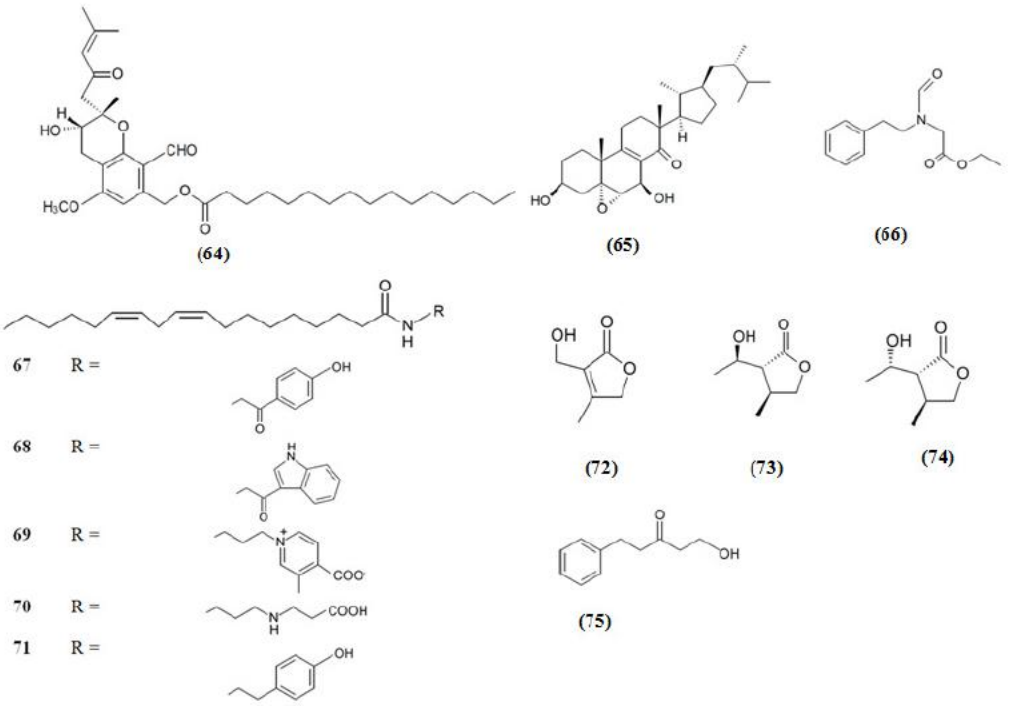


No	Termitomycesphin	C7-C9(C19)	n
48	A		16
49	B		18
50	C		16
51	D		18
52	E		16
53	F		18

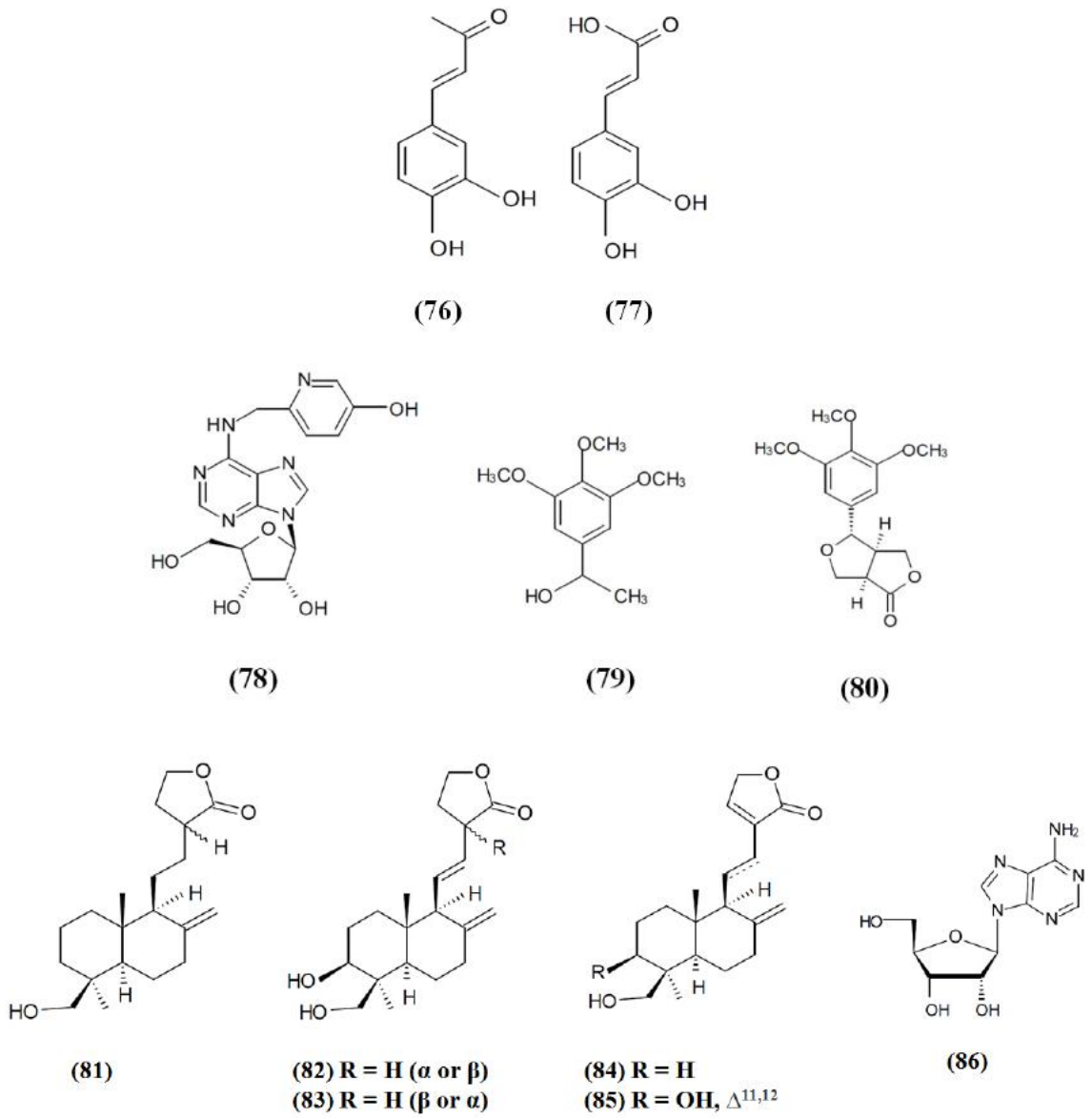


(54) n = 16
(55) n = 18

Supplemental Figure 4. Dictyophorines A **(43)** and B **(44)**, isolated from *Dictyophora indusiata*. Tricholomalides A-C **(45-47)** isolated from *Tricholoma* sp.; and Termitomycesphin A-H **(48-55)** isolated from *Termitomyces albuminosus*.



Supplemental Figure 5. 3-hydroxyhericenone F **(64)** isolated from *Hericium erinaceum;* strophasterol **(65)** from *Stropharia rugosoannulata;* Leccinine A **(66)** from *Leccinum extremiorientale*; termitomycamides A – E **(67-71)** from *Termitomyces titanicus*; and compounds **72-75** from *Mycoleptodonoides aitchisonii*.



Supplemental Figure 6. 3,4-dihydroxybenzalacetone **(76)** and caffeic acid **(77)** from *Inonotus obliquus.* N6-(5-hydroxy-2-pyridyl)-methyl-adenosine **(78)** from the mycelia of *A. mellea*; 1-(3,4,5-trimethoxyphenyl) ethanol **(79)** and caruilignan C **(80)** from *Daldinia concentric.* Neuroprotective diterpenes **(81-85)** and adenosine **(86)** from *Antrodia camphorate.*